CloudSage — Version Plan

Incremental versions from one EC2 service to a full optimization platform

V0 — AWS/EC2 Discovery

- EC2 metadata
- Basic CloudWatch retrieval
- Basic Cost Explorer retrieval
- CLI/API output
- No AI
- No writes

V1 — EC2 Cost Analyzer

- Normalized PostgreSQL data
- 7/30-day metrics
- Cost trends
- Deterministic candidate rules
- Savings calculator
- Basic API

V2 — AI Recommendation Engine

- Context builder
- Single LLM provider
- Structured output
- Evidence-based reasoning
- Risk/confidence fields
- AI output validation

V3 — Terraform Intelligence

- Terraform state mapping
- Terraform code/context
- Plan generation
- Plan JSON parsing
- Drift awareness
- No automatic apply

V4 — Human Approval

- Recommendation approval

- Plan approval
- Audit logs
- Policy engine
- Role separation

V5 — Controlled Apply

- Terraform apply
- Execution role
- Rollback strategy
- Failure handling
- Approval gates

V6 — Post-Optimization Verification

- Health baseline
- After-change metrics
- Cost comparison
- Savings realization
- Optimization success/failure classification

V7 — More AWS Services

- EBS
- Elastic IP
- S3
- RDS
- ALB
- ECS
- Lambda
- NAT Gateway

V8 — Dependency-Aware Optimization

- Resource relationships
- Application criticality
- Environment awareness
- SLA/SLO context
- Cross-resource recommendations

V9 — Production Platform

- Web UI
- Workers/queue

- Scheduling
- Multi-account support
- RBAC
- Secrets management
- Observability
- Notifications

V10 — Advanced Intelligence

- Historical optimization outcomes
- Recommendation quality metrics
- RAG over internal runbooks
- Natural-language queries
- Model evaluation suite
- Optional controlled automation

Versioning Principle

Each version must work before the next layer is introduced. The project should never simultaneously introduce a new AWS service, new AI architecture, new deployment platform and new automation model unless there is a specific reason.